

RED EARTH, AB, Canada

WMO: 712460

Lat: 56.5519N

Lon: 115.2778W

Elev: 546

StdP: 94.94

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.7	-32.1	-39.6	0.1	-35.6	-35.8	0.1	-32.1	7.4	-0.9	6.3	-3.6	1.0	150	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	28.1	16.9	26.2	15.8	24.4	15.0	18.4	24.9	17.2	23.5	16.2	22.2	3.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.2	20.6	14.9	11.3	19.3	13.9	10.6	18.2	54.3	24.8	50.5	23.7	47.5	22.3	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.9	6.0	5.3	DB	-38.7	31.1	3.5	1.6	-41.3	32.3	-43.3	33.2	-45.3	34.1	-47.9	35.2
			WB	-39.0	20.0	3.4	1.9	-41.4	21.3	-43.5	22.4	-45.4	23.5	-47.9	24.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.6	-14.4	-11.4	-5.7	3.0	9.6	14.0	16.3	14.7	9.6	3.0	-7.4	-13.3
	DBStd	12.68	9.50	7.98	8.34	5.46	4.72	3.35	2.90	3.38	4.31	4.56	7.22	8.84
	HDD10.0	3656	756	598	487	215	65	6	1	4	58	222	521	723
	HDD18.3	6146	1014	832	745	461	273	135	76	118	263	476	771	981
	CDD10.0	578	0	0	0	4	51	126	196	152	45	4	0	0
	CDD18.3	26	0	0	0	0	1	4	13	7	1	0	0	0
	CDH23.3	614	0	0	0	3	49	128	243	156	34	1	0	0
	CDH26.7	123	0	0	0	0	10	21	52	34	6	0	0	0
Wind	WSAvg	2.3	2.0	2.1	2.5	2.8	2.7	2.3	2.2	2.1	2.3	2.5	2.1	2.0
Precipitation	PrecAvg	486	27	18	21	24	46	88	85	57	41	28	27	24
	PrecMax	651	59	31	44	45	90	158	147	107	107	65	57	58
	PrecMin	317	7	6	5	4	15	30	47	19	4	13	11	7
	PrecStd	78	13	7	8	9	19	34	28	21	18	12	12	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	8.5	13.3	22.0	27.9	29.1	30.4	30.1	27.4	20.9	10.7	7.0
		MCWB	3.5	3.3	5.3	10.1	13.8	16.1	18.5	18.2	16.3	11.8	5.8	2.7
	2%	DB	4.6	5.7	10.0	17.7	24.2	26.6	28.0	27.2	23.5	15.7	7.1	4.1
		MCWB	2.0	1.7	3.9	7.8	12.2	14.8	17.6	17.2	14.6	8.7	3.3	1.4
	5%	DB	2.2	3.2	8.0	15.1	22.0	24.6	26.1	24.8	20.6	12.9	4.6	1.7
		MCWB	0.3	0.1	2.8	6.5	11.3	14.1	16.6	16.3	12.9	7.0	1.8	-0.4
	10%	DB	-0.5	0.7	5.9	12.6	19.7	22.6	24.1	22.7	17.9	10.7	2.3	-0.7
		MCWB	-2.0	-1.5	1.5	5.2	10.1	13.4	15.7	15.3	11.6	5.7	0.4	-2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	3.8	6.0	10.5	15.5	17.9	20.5	20.4	17.0	12.3	6.2	3.2
		MCDB	6.9	7.4	12.1	20.7	23.6	24.9	27.5	27.0	25.4	19.5	10.4	6.2
	2%	WB	2.2	1.9	4.4	8.4	13.4	16.4	18.8	18.2	15.2	9.4	3.7	1.6
		MCDB	4.4	5.0	9.3	16.3	22.3	23.7	25.4	24.5	21.5	14.8	6.6	4.1
	5%	WB	0.3	0.2	3.1	7.0	12.0	15.3	17.6	17.0	13.7	7.6	2.0	-0.2
		MCDB	2.2	3.2	7.5	14.2	20.2	21.8	23.7	23.1	19.5	11.8	4.2	1.6
	10%	WB	-2.0	-1.5	1.7	5.6	10.7	14.3	16.5	16.0	12.2	6.2	0.5	-2.1
		MCDB	-0.6	0.7	5.6	11.9	18.1	20.7	22.3	21.4	16.7	10.1	2.1	-0.7
Mean Daily Temperature Range	5% DB	MDBR	10.9	13.6	13.4	12.8	14.1	13.2	12.9	13.0	12.6	10.2	9.2	10.1
		MCDBR	11.9	16.0	14.3	17.3	18.1	17.6	17.0	16.9	17.6	14.7	11.1	10.9
		MCWBR	9.5	12.3	9.1	9.5	8.5	7.7	7.7	8.2	9.3	8.7	8.0	8.8
	5% WB	MCDBR	11.7	15.2	13.9	16.2	16.4	14.6	14.8	15.2	16.5	13.7	10.3	10.9
		MCWBR	9.4	11.9	9.1	9.2	8.0	7.0	7.5	8.0	9.1	8.5	7.7	9.0
Clear-Sky Solar Irradiance	taub	0.226	0.244	0.271	0.326	0.342	0.352	0.358	0.348	0.311	0.283	0.241	0.218	
	taud	2.312	2.356	2.391	2.394	2.431	2.431	2.432	2.436	2.558	2.592	2.443	2.238	
	Ebn at Noon	717	850	907	894	898	890	875	862	856	802	703	617	
	Edh at Noon	46	68	86	101	104	106	103	96	74	55	41	38	
All-Sky Solar Radiation	RadAvg	0.69	1.67	3.21	4.71	5.60	5.73	5.66	4.62	3.17	1.70	0.73	0.44	
	RadStd	0.06	0.14	0.25	0.40	0.49	0.46	0.34	0.35	0.26	0.24	0.07	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	+2.28	+3.22	N/A	N/A	N/A	N/A	N/A	+92	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+56	N/A

Nomenclature: See separate page